

Exercise 3 PL/SQL Sanskar Kumawat

Part 1 – Scenario 1 - The bank needs to process monthly interest for all savings accounts.

```
CREATE TABLE SavingsAccounts (  
    AccountID INT PRIMARY KEY,  
    CustomerID INT,  
    Balance DECIMAL(10,2)  
);  
  
INSERT INTO SavingsAccounts VALUES  
(101,1,10000), (102,2,15000), (103,3,20000), (104,4,8000), (105,5,12000);  
  
DELIMITER //  
CREATE PROCEDURE ProcessMonthlyInterest()  
BEGIN  
    UPDATE SavingsAccounts  
    SET Balance = Balance + (Balance * 0.01);  
END //  
DELIMITER ;  
  
CALL ProcessMonthlyInterest();  
SELECT * FROM SavingsAccounts;
```

Result Grid			
Filter Rows:			
	AccountID	CustomerID	Balance
▶	101	1	10100.00
	102	2	15150.00
	103	3	20200.00
	104	4	8080.00
	105	5	12120.00
*	NULL	NULL	NULL

Part 2 – Scenario 2 - The bank wants to implement a bonus scheme for employees based on their performance.

```
65 • CREATE TABLE Employees (  
66     EmployeeID INT PRIMARY KEY,  
67     EmployeeName VARCHAR(50),  
68     Department VARCHAR(50),  
69     Salary DECIMAL(10,2)  
70 );  
71 • INSERT INTO Employees VALUES  
72     (1,'Rahul','IT',50000), (2,'Amit','IT',60000), (3,'Priya','HR',45000),  
73     (4,'Sneha','Finance',55000),(5,'Rohan','IT',70000);  
74  
75     DELIMITER //  
76 • CREATE PROCEDURE UpdateEmployeeBonus(  
77     IN dept VARCHAR(50),  
78     IN bonusPercent DECIMAL(5,2)  
79 )  
80 BEGIN  
81     UPDATE Employees  
82     SET Salary = Salary + (Salary * bonusPercent / 100)  
83     WHERE Department = dept;  
84 END //  
85 DELIMITER ;  
86 • CALL UpdateEmployeeBonus('IT',10);  
87 SELECT * FROM Employees;
```

Result Grid		Filter Rows:		Edit:
	EmployeeID	EmployeeName	Department	Salary
▶	1	Rahul	IT	55000.00
	2	Amit	IT	66000.00
	3	Priya	HR	45000.00
	4	Sneha	Finance	55000.00
	5	Rohan	IT	77000.00
*	NULL	NULL	NULL	NULL

Part 3 - Scenario 3: Customers should be able to transfer funds between their accounts.

```
DELIMITER //
```

```
CREATE PROCEDURE TransferFunds(  
    IN fromAccount INT,  
    IN toAccount INT,  
    IN amount DECIMAL(10,2)  
)  
BEGIN  
    DECLARE currentBalance DECIMAL(10,2);  
    SELECT Balance  
    INTO currentBalance  
    FROM Accounts  
    WHERE AccountID = fromAccount;  
  
    IF currentBalance >= amount THEN  
        UPDATE Accounts  
        SET Balance = Balance - amount  
        WHERE AccountID = fromAccount;  
  
        UPDATE Accounts  
        SET Balance = Balance + amount  
        WHERE AccountID = toAccount;  
        SELECT 'Transfer Successful' AS Message;  
    ELSE  
        SELECT 'Insufficient Balance' AS Message;  
    END IF;  
END //
```

```
DELIMITER ;
```

Result Grid			
Filter Rows:			
	AccountID	CustomerName	Balance
▶	1001	Rahul	20000.00
	1002	Priya	20000.00
	1003	Amit	30000.00
*	NULL	NULL	NULL